

ANDREA DE PISAPIA

Control & Embedded Systems Engineer

PROFILE

Control & Embedded Systems Engineer with hands-on experience in FPGA-based real-time architectures, automation systems, and safety-critical control applications within the Fusion for Energy / ITER ecosystem.

Currently contributing to the development of the Optical Pellet Detection (OPD) system for the ITER Disruption Mitigation System (DMS), working on high-speed optical communication, FPGA firmware design, diagnostics, and distributed control architectures based on HotRIO and CompactRIO platforms.

Strong background in advanced control systems, embedded real-time applications, system integration, and robotics, with experience in multidisciplinary and international engineering environments.

WORK EXPERIENCE

Control & Embedded Systems Engineer (Trainee)

Fusion for Energy 

 Oct 2024 – Present

 Barcelona, Spain

- Contributed to the development of the FPGA firmware for the Optical Pellet Detection (OPD) system within the ITER Disruption Mitigation System (DMS), based on HotRIO architecture and integrated into the APDCAM/CompactRIO framework.
- Designed and integrated VHDL modules for high-speed optical communication using the proprietary HotRIO protocol on Lattice ECP5-5G FPGAs, including packet handling, clock synchronization, logical voting, and system diagnostics.
- Developed and documented multiple Finite State Machines (FSMs) for OPD startup and safety logic, implementing latch, trigger, reset, fault management, and PSOS operational sequences.
- Defined and maintained FPGA register maps through JSON-based register trees, while producing technical documentation and developing validation procedures for SFP-based HotRIO communication testing.
- Performed FPGA debugging, hardware/software integration, logic simulation, and validation of signals acquired from APDCAM systems and analog/digital sensors.
- Developed FPGA-based real-time signal processing architectures for high-speed ADC acquisition in the HotRIO control system for ITER Gyrotron Plants.
- Integrated FPGA firmware with CompactRIO systems, improving low-latency communication reliability over optical fiber links for distributed control applications.
- Collaborated with multidisciplinary teams in design reviews, debugging activities, and system-level validation for safety-critical control infrastructures.

PROJECTS

- Developed and validated MPC, RMPC/SMPC, and nonlinear control (Sliding Mode) for dynamic systems under uncertainty, with simulation-based verification (Monte Carlo methods).
- Implemented industrial control systems in Simulink, including motor drive control and closed-loop thermo-hydraulic regulation.
- Designed real-time embedded systems using FreeRTOS, including fault detection (watchdog-based) and deterministic control loops.
- Developed robotic control and motion planning for SCARA and mobile robots using ROS-based architectures and vision markers (ArUco).
- Worked on system-level modeling and multidisciplinary design optimization using virtual prototyping and RFLP methodologies.

CONTACT INFORMATION

 andreadepisapia3@gmail.com

 [linkedin.com](https://www.linkedin.com/in/andreadepisapia3/)

 +39 3338756044

 [github.com](https://github.com/andreadepisapia3/)

 Barcelona, ES

SKILLS

- Control & Simulation:**
MATLAB, Simulink, Simscape
- Embedded & Hardware:** FPGA (VHDL, Lattice Diamond), real-time communication systems
- Real-Time Systems:**
FreeRTOS
- Robotics:**
ROS, Gazebo
- Programming:**
C, C++, Python
- Other:**
CATIA, Xpress, Linux, Ubuntu

LANGUAGES

 Italian (Native Speaker)

 English (C1)

 Spanish (B1)

EDUCATION

Master of Automation Engineering and Robotics - 110/110

University of Napoli Federico II

 A.A: 2021/22 -> 2023/24

 Napoli, NA

 Thesis Title:
Development and Implementation of the HotRIO Control System for ITER Gyrotron Plants

Bachelor of Automation Engineering - 101/110

University of Napoli Federico II

 A.A: 2017/18 -> 2020/21

 Napoli, NA

 Thesis Title:
Arduino Controlled Rover Project